32.2: The history of animals spans more than half a billion years

To date, biologists have identified 1.3 million extant species of animals, and estimates of the actual number run far higher. This vast diversity encompasses a spectacular range of morphological variation, from corals to cockroaches to crocodiles. Various studies suggest that this great diversity originated during the last billion years. For example, researchers have unearthed 710-million-year-old sediments containing chemical evidence of steroids that today are primarily produced by a particular group of sponges. Since sponges are animals, these “fossil steroids” suggest that animals had arisen by 710 million years ago.

DNA analyses generally agree with this fossil biochemical evidence; for example, one recent molecular clock study estimated that sponges originated about 700 million years ago. These findings are also consistent with molecular analyses suggesting that the common ancestor of all extant animal species lived about 770 million years ago. What was this common ancestor like, and how did animals arise from their single-celled ancestors?

Steps in the Origin of Multicellular Animals

One way to gather information about the origin of animals is to identify protist groups that are closely related to animals. As shown in **Figure 32.3**, morphological and molecular evidence point to choanoflagellates as the closest living relatives of animals. Based on such evidence, researchers have hypothesized that the common ancestor of choanoflagellates and living animals may have been a **suspension feeder** similar to present-day choanoflagellates.

Figure 32.3

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    Three lines of evidence that choanoflagellates are closely
related to animals.
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Three lines of evidence that choanoflagellates are closely related to animals.

1. Morphologically, choanoflagellate cells and the collar cells (or choanocytes) of sponges, a basal group of animals, are almost indistinguishable.
2. Similar collar cells have been identified in other animals, including cnidarians, flatworms, and echinoderms - but they have never been observed in non-choanoflagellate protists or in plants or

fungi.

3. D N A sequence data indicate that choanoflagellates and animals are sister groups. In addition, genes for signaling and adhesion proteins previously known only from animals have been discovered in choanoflagellates. On an evolutionary tree, the D N A sequence data are marked at the common ancestor of choanoflagellates and animals.

Are the data described in **3** *consistent with predictions that could be made from the evidence in* **1** *and* **2**? *Explain.*

*For suggested answer, see*Appendix A.*

Scientists exploring *how* animals may have arisen from their single-celled ancestors have noted that the origin of multicellularity requires the evolution of new ways for cells to adhere (attach) and signal (communicate) to each other. To learn more about such mechanisms, researchers compared the genome of the unicellular choanoflagellate *Monosiga brevicollis* with those of representative animals. This analysis uncovered 78 protein domains in *M. brevicollis* that were otherwise only known to occur in animals. (A *domain* is a key structural or functional region of a protein.) For example, *M. brevicollis* has genes that encode domains of certain proteins (known as cadherins) that play key roles in how animal cells attach to one another, as well as genes that encode protein domains that animals (and only animals) use in cell-signaling pathways.

Let's take a closer look at the cadherin attachment proteins we just mentioned. DNA sequence analyses show that animal cadherin proteins are composed primarily of domains that are also found in a cadherin-like protein of choanoflagellates **(Figure 32.4)**. However, animal cadherin proteins also contain a highly conserved region not found in the choanoflagellate protein (the "CCD" domain labeled in Figure 32.4). These data suggest that the cadherin attachment protein originated by the rearrangement of protein domains found in choanoflagellates plus the incorporation of a novel domain, the conserved CCD region. Overall, comparisons of choanoflagellate and animal genomes suggest that key steps in the transition to multicellularity in animals involved new ways of using proteins or parts of proteins that were encoded by genes found in choanoflagellates.

Figure 32.4

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Cadherin proteins in choanoflagellates and animals.
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On a tree, a branch splits to choanoflagellate and animals. A protein is compared between the groups. In animals, the C C D domain of the protein is found.

The ancestral cadherin-like protein of choanoflagellates has seven kinds of domains (regions), each represented here by a particular symbol. With the exception of the "CCD" domain, which is found only in animals, the domains of animal cadherin proteins are present in the choanoflagellate cadherin-like protein. The cadherin protein domains shown here were identified from whole-genome sequence data; evolutionary relationships are based on morphological and DNA sequence data.

Next, we'll survey the fossil evidence for how animals evolved from their distant common ancestor over four geologic eras (see Table 25.1 to review the geologic time scale).

Interview

Interview with Nicole King: Investigating the ancestry of choanoflagellates

Nicole King.

Neoproterozoic Era (1 Billion–541 Million Years Ago)

Although data from fossil steroids and molecular clocks indicate an earlier origin, the first generally accepted macroscopic fossils of animals date from about 560 million years ago. These fossils are members of an early group of mostly soft-bodied multicellular eukaryotes, known collectively as the **Ediacaran biota**. The name comes from the Ediacara Hills of Australia, where fossils of these organisms were first discovered **(Figure 32.5)**. Similar fossils have since been found on other continents. Among the oldest Ediacaran fossils that resemble animals, some have been classified as molluscs (snails and their relatives) or close relatives of the molluscs, while others are thought to be sponges or cnidarians (sea anemones and their relatives). Still others have proved difficult to classify, as they do not seem to be closely related to any living animal or algal groups. In addition to these macroscopic fossils, Neoproterozoic rocks have also yielded what may be microscopic fossils of early animal embryos. Although these microfossils appear to exhibit the basic structural organization of present-day animal embryos, debate continues about whether these fossils are indeed of animals.

Figure 32.5

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Ediacaran fossil animals.
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Two fossils. Part a. Dickinsonia costata (taxonomic affiliation unknown). Scale: 2.5 centimeters. Part b. Kimberella, a mollusk (or close relative). Scale: 1 cm.

Fossils dating to 560 million years ago document the earliest known macroscopic animals, including these two species.

The fossil record from the Ediacaran period (635–541 million years ago) also provides early evidence of predation. Consider *Cloudina*, a small animal whose body was protected by a shell resembling a series of nested cones **(Figure 32.6)**. Some *Cloudina* fossils show signs of attack: round "bore holes" that resemble those formed today by predators that drill through the shells of their prey to gain access to the soft-bodied organisms lying within. Like *Cloudina*, some other small Ediacaran animals had shells or other defensive structures that may have been selected for by predators. Overall, the fossil evidence indicates that the Ediacaran was a time of increasing animal diversity—a trend that continued in the Paleozoic.

Figure 32.6

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Early evidence of predation.
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Fossil of the animal Cloudina with a bore hole through its shell. Scale: 0.1 millimeters.

This 550-million-year-old fossil of the animal *Cloudina* shows evidence of having been attacked by a predator that bored through its shell.

Paleozoic Era (541–252 Million Years Ago)

Another wave of animal diversification occurred 535–525 million years ago, early in the Cambrian period of the Paleozoic era—a phenomenon referred to as the **Cambrian explosion** (see Concept 25.3). In strata formed before the Cambrian explosion, only a few animal phyla have been observed. But in strata that are 535–525 million years old, paleontologists have found the oldest fossils of about half of all extant animal phyla, including the first arthropods, chordates, and echinoderms. Many of these fossils, which include the first large animals with hard, mineralized skeletons, look very different from most living animals **(Figure 32.7)**. Even so, paleontologists have established that these Cambrian fossils are members of extant animal phyla, or at least are close relatives. In particular, most of the fossils from the Cambrian explosion are of **bilaterians**, an enormous clade whose members (unlike sponges and cnidarians) typically have a two-sided or bilaterally symmetric form and a complete digestive tract, an efficient digestive system that has a mouth at one end and an anus at the other. As we'll discuss later in the chapter, bilaterians include molluscs, arthropods, chordates, and most other living animal phyla.

Figure 32.7

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A Cambrian seascape.
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Hallucigenia fossil (530 million years ago). Scale: 1 centimeter.

This artist's reconstruction depicts a diverse array of organisms found in fossils from the Burgess Shale site in British Columbia, Canada. The animals include Pikaia (eel-like chordate at top left), Marella (small arthropod swimming at left), Anomalocaris (large animal with grasping limbs near a circular mouth), and Hallucigenia (animals with toothpick-like spikes on the seafloor and in inset).

As the diversity of animal phyla increased during the Cambrian, the diversity of Ediacaran life-forms declined. What caused these trends? Fossil evidence suggests that during the early Cambrian period, as predators acquired novel adaptations for catching prey, such as claws, prey species acquired new defenses, such as protective shells. As new predator-prey relationships emerged, natural selection may have led to the decline of the soft-bodied Ediacaran species and the rise of various bilaterian phyla. Another hypothesis focuses on an increase in atmospheric oxygen that preceded the Cambrian explosion. More plentiful oxygen would have enabled animals with higher metabolic rates and larger body sizes to thrive, while potentially harming other species. A third hypothesis proposes that genetic changes affecting development, such as the origin of *Hox* genes and the addition of new microRNAs (small RNAs involved in gene regulation), facilitated the evolution of new body forms. In the **Scientific Skills Exercise**, you can investigate whether there is a correlation between microRNAs (miRNAs; see Figure 18.15) and body complexity in various animal phyla. These various hypotheses are not mutually exclusive; predator-prey relationships, atmospheric changes, and changes in development may each have played a role.

Scientific Skills Exercise

Calculating and Interpreting Correlation Coefficients

AP Science Practices 4, 5, 6

Is Animal Complexity Correlated with miRNA Diversity? Animal phyla vary greatly in morphology, from simple sponges that lack tissues and symmetry to complex vertebrates. Members of different animal phyla have similar developmental genes, but the number of miRNAs varies considerably. In this exercise, you will explore whether miRNA diversity is correlated to morphological complexity.

How the Study Was Done In the analysis, miRNA diversity is represented by the average number of miRNAs in a phylum (x), while morphological complexity is represented by the average number of cell types

(y). Researchers examined the relationship between these two variables by calculating the correlation coefficient (r). The correlation coefficient indicates the extent and direction of a linear relationship between two variables (x and y) and ranges in value between −1 and 1. When r<0, x and y are negatively correlated, meaning that in a plot of the data points, values of y become smaller as values of x become larger. When r>0, x and y are positively correlated (y becomes larger as x becomes larger). When r=0, the variables are not correlated.

The formula for the correlation coefficient r is the following:

$r=\sum(x_i-\bar{x})(y_i-\bar{y})(n-1)(s_x s_y)$

In this formula, n is the number of observations, x_i is the value of the ith observation of variable x, and y_i is the value of the ith observation of variable y. $\bar{x}$ and $\bar{y}$ are the means of variables x and y, and s_x and s_y are the standard deviations of variables x and y. The Σ symbol indicates that the n values of the product $(x_i-\bar{x})(y_i-\bar{y})$ are to be added together.

Data from the Study

A table lists animal phylum, number of m i R N As, and number of cell types.

Data from B. Deline et al., Evolution of metazoan morphological disparity, *Proceedings of the National Academy of Sciences USA* 115:E8909–E8918 (2018).

The table is as follows:

Animal phylum	i	Number of m i R N A's (x sub i)	left parenthesis, x sub i minus x bar, right parenthesis	left parenthesis, x sub i minus x bar, right parenthesis, squared	Number of cell types (y sub i)	left parenthesis, y sub i minus y bar, right parenthesis	pa
Porifera	1	5.8			25		
Platyhelminthes		2	35			30	

Cnidaria	3	2.5			34		
Nematoda	4	26			38		
Echinodermata	5	38.6			45		
Cephalochordata	6	33			68		
Arthropoda	7	59.1			73		
Urochordata	8	25			77		
Mollusca	9	50.8			83		
Annelida	10	58			94		
Vertebrata	11	147.5			172.5		
		x bar = ; s sub x =		Sum of equals	y bar = ; s sub y =		Su

INTERPRET THE DATA

- First, practice reading the data table. For the eighth observation i=8, what are xi and yi? For which phylum are these data?
- Next, we'll calculate the mean and standard deviation for each variable (to review these topics, see the Scientific Skills Review in Appendix D).
-
    - The **mean** (x^{-}) is the sum of the data values divided by n, the number of observations: $x^{-}=\Sigma xin$ Calculate the mean number of miRNAs (x^{-}) and the mean number of cell types (y^{-}) and enter them in the data table (to calculate y^{-}, replace each x in the formula with a y).
- Next, calculate ($xi-x^{-}$) and ($yi-y^{-}$) for each observation, recording your results in the appropriate column. Square each of those results to complete the $(xi-x^{-})2$ and $(yi-y^{-})2$ columns; sum the results for those columns.
- Next, we'll calculate the **standard deviation**, s, which describes the variation found in a data set. Calculate sx, the standard deviation of the miRNA data, using the following formula, where n=the number of observations. [Use the sum of $(xi-x^{-})2$ you found in part (b).]

$sx=\Sigma(xi-x^{-})2n-1$

- Repeat to find sy by substituting y for x in this equation.
- Next, we'll calculate the correlation coefficient r for the variables x and y.
-
    - First, use the results from question 2(b) to complete the $(x_i - \bar{x})(y_i - \bar{y})$ column; sum the results in that column.
- Now find the product of sxsy using your results from 2(c) above. Plug these values into the formula for r.
- Do these data indicate that miRNA diversity and animal complexity are negatively correlated, positively correlated, or uncorrelated? Explain.
- What does your analysis suggest about the role of miRNA diversity in the evolution of animal complexity?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

The Cambrian period was followed by the Ordovician, Silurian, and Devonian periods, when animal diversity continued to increase, although punctuated by episodes of mass extinction (see Figure 25.18). Vertebrates (fishes) emerged as the top predators of the marine food web. By 450 million years ago, groups that diversified during the Cambrian period began to make an impact on land. Arthropods were the first animals to adapt to terrestrial habitats, as indicated by fragments of arthropod remains and by well-preserved fossils from several continents of millipedes, centipedes, and spiders. Another clue is seen in fossilized fern galls—enlarged cavities that fern plants form in response to stimulation by resident insects, which then use the galls for protection. Fossils indicate that fern galls date back at least 302 million years, suggesting that insects and plants were influencing each other's evolution by that time.

Vertebrates colonized land around 365 million years ago and diversified into numerous terrestrial groups. Two of these survive today: the amphibians (such as frogs and salamanders) and the amniotes (reptiles, including birds, and mammals). We will explore these groups, known collectively as the tetrapods, in more detail in Chapter 34.

Mesozoic Era (252–66 Million Years Ago)

The animal phyla that had evolved during the Paleozoic began to spread into new habitats. In the oceans, the first coral reefs formed, providing other marine animals with new places to live. Some reptiles returned to the water, leaving plesiosaurs (see Figure 34.25) and other large aquatic predators as their descendants. On land, descent with modification in some tetrapods led to the origin of wings and other flight equipment in pterosaurs and birds. Large and small dinosaurs emerged, as both predators and herbivores. At the same

time, the first mammals—tiny nocturnal insect-eaters—appeared on the scene. In addition, as you read in Concept 30.3, flowering plants (angiosperms) and insects both underwent dramatic diversifications during the late Mesozoic.

Cenozoic Era (66 Million Years Ago to the Present)

Mass extinctions of both terrestrial and marine animals ushered in a new era, the Cenozoic. Among the groups of species that disappeared were the large, nonflying dinosaurs and the marine reptiles. The fossil record of the early Cenozoic documents the rise of large mammalian herbivores and predators as mammals began to exploit the vacated ecological niches. The global climate gradually cooled throughout the Cenozoic, triggering significant shifts in many animal lineages. Among primates, for example, some species in Africa adapted to the open woodlands and savannas that replaced many of the former dense forests. The ancestors of our own species were among those grassland apes.